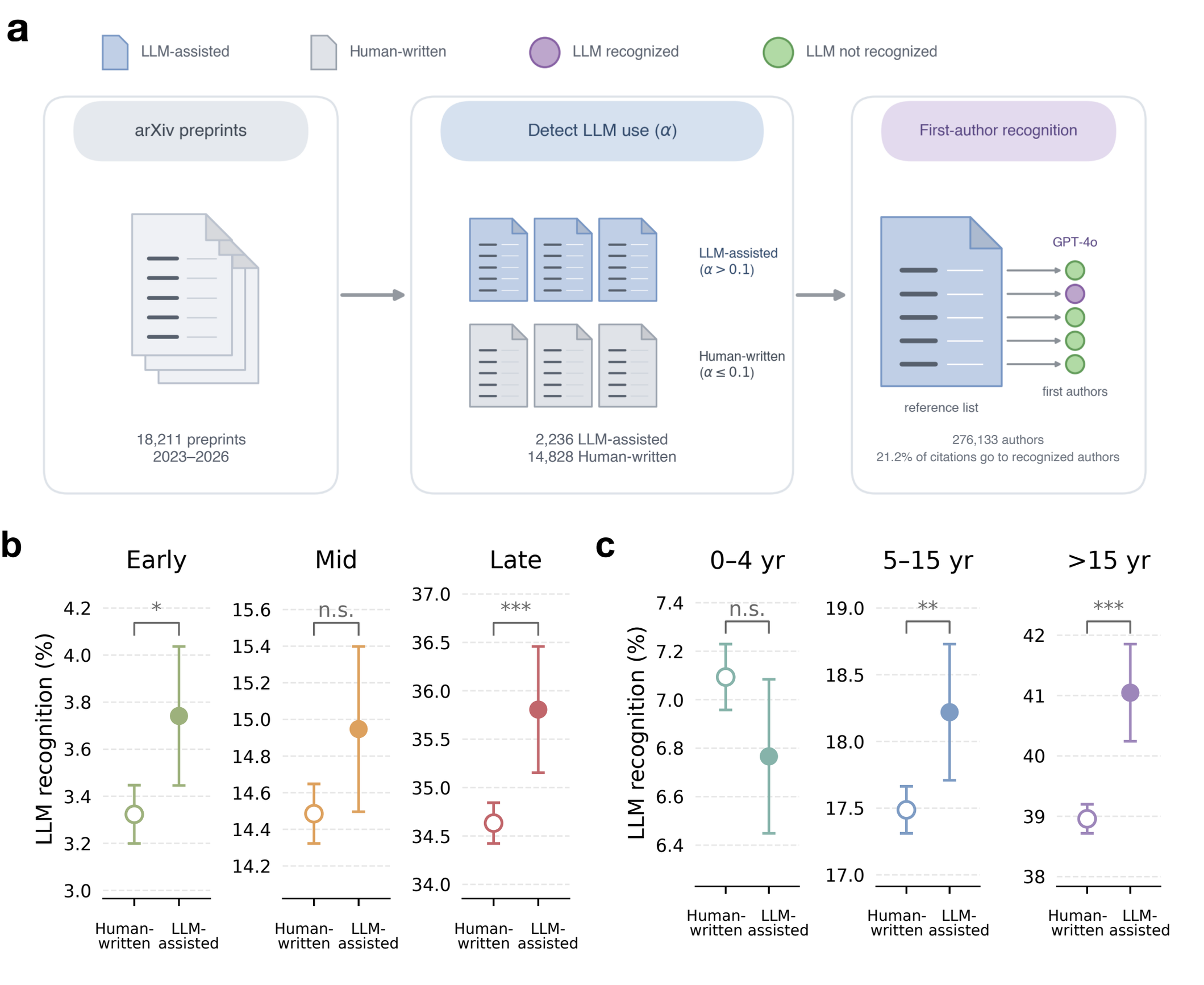

a
LLM-assisted
Human-written
LLM recognized
LLM not recognized
arXiv preprints
18,211 preprints
2023–2026
Detect LLM use (α)
LLM-assisted
(α > 0.1)
Human-written
(α ≤ 0.1)
2,236 LLM-assisted
14,828 Human-written
First-author recognition
GPT-4o
first authors
reference list
276,133 authors
21.2% of citations go to recognized authors
b
Early
Mid
Late
LLM recognition (%)
4.2
4.0
3.8
3.6
3.4
3.2
3.0
15.6
15.4
15.2
15.0
14.8
14.6
14.4
14.2
37.0
36.5
36.0
35.5
35.0
34.5
34.0
*
n.s.
***
Human-written
LLM-assisted
c
0–4 yr
5–15 yr
>15 yr
LLM recognition (%)
7.4
7.2
7.0
6.8
6.6
6.4
19.0
18.5
18.0
17.5
17.0
42
41
40
39
38
n.s.
**
***
Human-written
LLM-assisted